

## Post-Trade Analysis

Algorithmic post-trade analysis is a two part process that consists of cost measurement and algorithm performance analysis. First, cost is measured as the difference between the actual realized execution price and the specified benchmark price. This allows investors to critique the accuracy of the trading cost model to improve future cost estimates and macro strategy decisions, and provide managers with higher quality price information to improve investment decisions. Second, algorithmic performance is analyzed to assess the ability of the algorithm to adhere to the optimally prescribed strategy, its ability to achieve fair and reasonable prices, and determine if the algorithm deviates from the optimally specified strategy in an appropriate manner. Investors must continuously perform post-trade analysis to ensure brokers are delivering as advertised, and question those executions that are out of line with pre-trade cost estimates.

## Rule-Based Trading

Rule-based trading algorithms control the macro-level decisions such as order slicing strategies that break larger orders into smaller pieces to trade over time in order to reduce market impact cost. These instructions, however, are based on simple logic and heuristics, with many of the guidelines being completely arbitrary (e.g., participate with volume over a specified period, trade faster when prices are favorable, trade slower in times of adverse trends, etc.). Furthermore, many of these so-called rule-based algorithms do not provide insight into potential costs or associated risk. Furthermore, they do not provide necessary transparency to determine the most appropriate algorithm given the objectives of the fund. For the most part, rule-based trading algorithms are a “black box” approach to trading.

## Quantitative Techniques

Quantitative algorithmic trading also controls the macro-level decisions but differs from rule-based trading in that all decisions are based on robust statistical models and a sound mathematical framework. For example, quantitative models serve as the basis for developing strategies to minimize the difference between expected execution price and a specified benchmark price, to minimize cost subject to a maximum level of risk exposure, or to maximize the probability of transacting more favorably than a specified benchmark price. Quantitative algorithms manage overall transaction costs (e.g., market impact, price momentum, and timing risk), and provide investors with necessary transparency regarding cost and